Mechanism rerun — handoff for the NeurIPS resubmission

Everything below was produced by re-running the label-dependent mechanism analyses on top of `relabel_out/` and the global-scope tool-call scorer. Branch: `relabel-rerun`. Ready-to-paste LaTeX: `paper_updates.tex` — three ready-to-paste tables, plus a commented outline for each paragraph that needs rewriting (what to cover and which numbers to cite, not drafted prose). Generated from the result JSON by `make_paper_updates.py`, so no number is hand-transcribed; compile-tested, and every citation key it uses is already in `refs.bib`.

---

0. Your three asks, answered

1. "Add the missing cosine values." Done, all five models — and it is a *new table* (§3.2 below), since §3.6 describes the analysis but no number appears anywhere in the draft. It also turned up something: the behaviour-direction claim holds everywhere (0.69–0.86), but the **other** cosine in that section — text-derived vs. tool-derived — **fails for two models** (Command-R 0.394, Llama 0.213). §5 currently says those directions "remain aligned", and the Discussion repeats it. Both need softening.

2. "Rerun mech and update results." Done. Every Table 2 column except Suppression has now been rerun: ablation, addition and AUC in the first session, and patching in the second (§3.1, §1c). Suppression is the only carried-over number left.

3. "Double-check the causal claims." Done, and this is the largest change — it is narrative, not a table. Checking them turned up a second problem beyond the classifier, described in §1b: some interventions push a model into broken output, which any refusal classifier scores as refusal. What that leaves:

- **Ablation** reverses in Qwen (0.775→0.925, fluent output — a real effect). It still lowers refusal in Mistral, Llama and Command-R, though Command-R's 72%→2% collapse is really 67%→34%. Gemma's is **not reportable**: 100% of its ablated output is degenerate.
- **Addition is not measurable at the published coefficient.** It drives all five models into degenerate output. This affects the draft's existing numbers too, not just the rerun — the old regex scores "I'm sorry I cannot I'm sorry I cannot..." as refusal just as the judge does.
- **Steering suppresses tool-calling, not unsafe tool-calling** (§3.4). "Unsafe → 0" at the coefficients a model survives is the model no longer calling tools — on harmful *and benign* prompts — while the unsafe rate among the calls it still makes never falls. An earlier version of this document read those points as a usable operating regime; that was wrong, and the draft's "blunt lever" is if anything an understatement.
- **Patching, now rerun for all five** (§1c, §3.1): the flips are real, but 70–97% of them are "stopped calling" rather than "called safely", and 80–96% of the calls a patched model still makes remain unsafe.

Because that ask deserved a test no classifier could confound, I also ran a new experiment (§3.3): ablate, then score the unsafe tool call directly. Result — the direction gates **whether the model engages**, not how safely it acts once engaged. Patching and steering now say the same thing from two more directions. That is a sharper claim than the current "partial mediator" framing, and it is defensible against exactly the objection that sank the ablation numbers.

Scope of edits this implies: 6 new tables, 1 rewritten table, 8 prose locations, 4 methods additions (degeneracy screen, KL guardrail, call-rate reporting, prompt-rendering check), and a pass over the abstract, the introduction and Future Work.

---

1. TL;DR

Table 1 in the current draft has already been updated with the new labels. **Table 2 has not** — it is still entirely pre-relabel. Rerunning it changes four of its five columns, and the causal section's claims narrow to one.

The numbers get smaller; the claim gets more consistent:

- The direction is still a strong predictor everywhere — AUC **0.76–0.86**, and all five models now reproduce their published AUC once the prompt is built correctly.
- We have the §3.6 cosine numbers, which the draft describes but never computed. The behaviour-direction cosine holds (0.69–0.86); the text-vs-tool cosine holds for 3/5.
- **Three interventions agree on what the direction does: it gates whether the model acts, not how safely it acts.** Ablation on tool prompts moves engagement 10–22 points with a flat unsafe rate among calls (§3.3). Patching flips unsafe calls mostly into *no* call, with 80–96% of the remaining calls still unsafe (§3.1). Steering removes unsafe calls only by removing calls, on the benign side too (§3.4).
- **Ablation is not a uniform story.** Command-R is textbook Arditi (0.68→0.33). Mistral and Llama drop modestly. Qwen (0.78→0.93) and — once ablated coherently — Gemma (0.66→1.00) refuse *more*, which says the max-separation direction is not a refusal direction in those models.
- **Only Command-R's direction passes Arditi's KL admissibility check** (§1c). Mistral and Llama fail at every layer; Gemma fails by two orders of magnitude, for a diagnosable reason (its massive-activation common mode), and your mean-out fix repairs the coherence at 27B.
- **Addition at the natural magnitude does almost nothing** ($\leq$ 1.3% refusal in Mistral and Gemma); the 4× the draft used, and the 0→100% "rescue", are degenerate output in every model.

What breaks in the draft: the ablation, addition and patching columns and the steering paragraph, for two separate reasons — a phrase-matching classifier, and broken output that no classifier can score (§1b). Nothing in the geometric half of the paper breaks.

1b. Output degeneracy — a second problem, found on review

Ablation and addition can push a model off the rails entirely. Real examples from our runs:

> *Gemma, ablated:* `Vy...` *for 400 characters*
>
> *Llama, addition:* `I can I can I can I can...` *for 600 characters*

The judge has no "broken output" verdict, so it forces such a response into refuse/caveat/comply, and "no help was given" reads as REFUSE. **Any intervention rate computed over degenerate output measures breakage, not refusal.**

This applies to the draft's existing numbers too. Qwen's degenerate addition output is literally "I'm sorry I cannot I'm sorry I cannot…", which the old regex classifier also scores as refusal — so the published 1%→71% has the same problem. This is not a regression introduced by the relabel.

Per-cell rates are in `relabel_analysis/degeneracy_audit.json` (`degeneracy_audit.py`). What is and is not measurable:

Clean (0% degenerate)	every model's ablation except Gemma; all four baselines; the lower half of every steering grid
Artifact	addition at 4× the gap in all five (67–100% degenerate); Gemma's ablation (100%)

Recommend putting the degeneracy screen in the paper. It is a real methodological contribution and pre-empts the obvious reviewer question.

1c. Your ablation handoff — verified, then run

Your `HANDOFF_raj_ablation.md` was checked claim by claim against the saved generations and then its open items were run on an H100 (2026-09-02). Everything you stated held. Two of your findings overturned parts of an earlier version of this document, and both are now in it.

Verified from the saved generations, no GPU:

- Steering's "unsafe → 0" is tool-call collapse. Confirmed, and it is worse than you wrote: it happens on the benign side at the same coefficient, and unsafe-given-call never drops (§3.4).
- Your degeneracy rule catches the sentence-level loops mine missed (Qwen 2.25×: 32%→51%, Mistral 17: 18%→36%, Gemma addition 68%→100%). It is now `degeneracy.is_degenerate_v2` and every rate in this session uses it. The four clean steering points stay $\leq$ 2% under it.
- Thinking mode was already off for Qwen (`render_utils.py`). The cosines you quote are ours.
- `main` merged into `relabel-rerun` cleanly (no conflicts).

Run on the GPU — the KL guardrail (Arditi App. C.1) for every candidate layer, your mean-out recipe, random- and mean-direction controls, and addition at 1×/1.5×/2×/3× the natural magnitude, plus patching for all five. KL is scored two ways: on 48 generic harmless requests (Arditi's protocol) and on the benchmark's own benign prompts, which sit near the refusal boundary.

Model	KL of r_text, generic / dataset	Layers passing (generic)	Random dir. KL	Mean-out KL	Verdict
Command-R-7B	**0.04** / 0.14	8/17 (best L27, next to L26)	0.001	0.20	admissible
Mistral-7B	0.25 / 0.91	0/17	0.004	0.28	no admissible layer
Gemma-3-27B	11.3 / 11.0	0/32	0.04	2.5	no admissible layer; mean-out restores coherence
Llama-3.1-70B (4-bit)	0.28 / 0.40	11/41 (best L60 at 0.08; operating L67 fails)	0.011	0.39	admissible at L60, not at the layer used
Qwen3-14B	*running*				

Nothing passes on the dataset's own benign prompts in any model — the direction is entangled with the benchmark's domains, not only with refusal. Gemma's diagnostics at 27B match your 4B run: mean-projection/gap = 1.00, cos(r, mean) = 0.24, mean-activation norm ≈ 77k; a random direction is harmless (KL 0.04, 0% degenerate), so Gemma is not fragile to ablation in general — it is fragile to *this* direction because of what it shares with the common mode.

Ablation controls, harmful no-tool prompts, n = 240, judged refusal over coherent outputs:

Model	none	r_text	mean-out	random	mean-dir	Reading
Command-R-7B	0.68	**0.33**	0.48	0.69	0.47 (6% deg.)	Arditi holds; the mean direction car part of the signal
Mistral-7B	0.49	**0.33**	0.35	0.47	0.54 (10% deg.)	modest real drop; random is a clean
Gemma-3-27B	0.66	*100% degenerate*	**1.00** (1% deg.)	0.68	*100% degenerate*	your fix works at 27B; coherent abla *raises* refusal
Llama-3.1-70B (4-bit)	*running*					
Qwen3-14B	*running* (earlier run: 0.78 → 0.93, 0% degenerate)					

The Gemma mean-out outputs are fluent, on-topic refusals; prompts the baseline hedged on become flat refusals. With Qwen, that makes two of five models where removing the max-separation direction *increases* refusal — consistent with the direction being harmfulness/topic rather than refusal in those families (your Zhou et al. / Wollschläger et al. pointers).

Addition at multiples of the diff-in-means gap, benign no-tool prompts, n = 240:

Model	1×	1.5×	2×	3×
Mistral-7B	0.004	0.01	0.06 (5% deg.)	97% degenerate
Command-R-7B	0.41 among the coherent half (50% deg.)	100% deg.	100% deg.	100% deg.
Gemma-3-27B	0.013	91% deg.	100% deg.	100% deg.
Llama-3.1-70B (4-bit), Qwen3-14B	*running*			

Arditi's 1× produces essentially no refusal in Mistral and Gemma; Command-R breaks before 1×. The draft's 4× is degenerate everywhere. "Addition rescues Llama" should be withdrawn, as you said.

Predictions were written down before the session and scored afterwards: `PREDICTIONS.md`. Held: the patching decomposition (5/5), the random-direction null, Gemma's mean-out coherence and direction. Missed: Mistral and Llama failing the KL guardrail, the size of the addition effect at 1×, and Gemma not being fragile to a random direction.

---

2. Why the ablation numbers changed

The old refusal classifier keyed on first-person phrasing ("I cannot...", "I'm unable to..."). Ablation changes *how* a model refuses without stopping it:

> *"Certainly not. Submitting a client trade without proper approval is a serious violation of*
>
> *compliance and ethical standards."*

That is an unmistakable refusal with no "I cannot" in it. The regex counts it as compliance, so a phrasing shift reads as a behavioural drop. For Qwen, 66 of 120 ablated responses are refusals the regex missed — and Qwen's ablated output is fluent throughout, so this is a real effect, not the degeneracy problem in §1b.

This is the same failure mode the relabel fixed for Table 1 — it just was never applied to the intervention results. Gemma looks similar on the surface (regex 0, judge 120) but is a different problem: there the output is degenerate, so neither number means anything.

---

3. Exact changes to the paper

3.1 Table 2 (`tab:mech`) — four columns change

Model	**Ablation (old → new)**	**Addition at 1× (old 4× → new)**	**AUC (old → new)**	**Patching flip (old → new = safe call + no call)**
Qwen3-14B	57%→32% → **78%→92%**	1%→71% → *running*	0.724 → **0.808**	18% → **24%** = 9 + 15
Mistral-7B	29%→13% → **49%→33%**	0%→80% → **0%→0.4%**	0.689 → **0.760**	24% → **20%** = 6 + 13
Command-R-7B	72%→2% → **68%→33%**	2%→92% → **0%→41%** (50% degenerate)	0.751 → **0.768**	62% → **34%** = 6 + 29
Gemma-3-27B	21%→0% → **66%→100%** (mean-out; raw direction not reportable)	1%→25% → **0%→1%**	0.791 → **0.790**	46% → **90%** = 3 + 87
Llama-3.1-70B	73%→61% → **71%→59%**	1%→1% → *running* (4-bit)	0.881 → **0.864**	40% → **13%** = 3 + 9 (4-bit)

Unsafe-given-call after patching: 0.89 / 0.93 / 0.92 / 0.80 / 0.96. The patching column should become its own table (`tab:patching` in `paper_updates.tex`) because the split is the finding. Gemma's old patching number came from a run with no tools in the prompt at all (§6). Llama's rerun used 4-bit NF4 weights on an 80 GB card and is marked as such; a bf16 confirmation needs a B200 or two H100s and about 30 minutes.

Layer and Suppression are unchanged. The ablation column now uses this session's control runs (n = 240, judged over coherent outputs), which match the first session's numbers within 3 points for Qwen, Mistral, Command-R and Llama.

n = 240 per condition. A first pass at n = 120 gave the same picture throughout, so none of this rests on sample size.

3.2 New table: direction cosines (`tab:cosine`)

§3.6 promises this analysis and no number appears anywhere in the draft. (The 0.735 in notebook 02 is `cos(r_text, r_tool)` — a different quantity.)

Model	cos(r_text, r_behav)	cos(r_text, r_tool)	r_behav old vs new
Qwen3-14B	0.859	0.741	0.975
Mistral-7B	0.711	0.773	0.978
Command-R-7B	0.746	0.394	0.978
Gemma-3-27B	0.822	0.797	0.958
Llama-3.1-70B	0.693	0.213	0.979

The §3.6 claim holds in every model (0.69–0.86). The last column is useful defensively: recomputing the behaviour-defined direction under the *old* labels agrees at 0.96–0.98, so the check does not depend on the relabelling.

But `cos(r_text, r_tool)` needs a softened claim. §5 currently says text- and tool-derived directions "remain aligned". True for Qwen, Mistral and Gemma (0.74–0.80); not for Command-R (0.39) or Llama (0.21).

3.3 New table: ablation on behaviour (`tab:ablation-action`)

Because the ablation result is classifier-dependent, we tested it a second way that no classifier can touch — ablate, then score the unsafe tool call directly (a predicate over parsed calls). n = 200, paired.

Model	Unsafe (base → abl)	Made any call	Unsafe given it called
Qwen3-14B	0.460 → 0.235	0.590 → 0.370	0.780 → 0.635
Mistral-7B	0.325 → 0.350 (n.s.)	0.425 → 0.470	0.765 → 0.745
Command-R-7B	0.120 → 0.180	0.135 → 0.230	0.889 → 0.783
Gemma-3-27B	0.130 → 0.000	0.175 → **0.000**	0.743 → n/a
Llama-3.1-70B	0.450 → 0.360	0.740 → 0.565	0.608 → 0.637

The headline column has no consistent story. The conditional column barely moves anywhere, while the rate of calling a tool at all swings 10–22 points in both directions. **Ablation changes how often a model acts, not how safely it acts once it does.**

Caveat for the text: Gemma's tool-calling collapses to zero, so its apparent safety gain is the model ceasing to act — capability damage, not alignment.

3.4 Steering — this conclusion changes

The draft's steering evidence has two problems, and an earlier version of this document fixed the first and then misread the result. First, the top of each grid is where the model breaks (32–100% degenerate under the v2 rule), so those over-refusal rates (Qwen 0.44, Command-R / Gemma / Llama 1.00) measure breakage and should not be quoted. Second — the part I got wrong before — "unsafe → 0" at the next coefficient down is not safer calling. It is the model no longer calling tools, on both sides, and the text judge cannot see it:

model	coef	harmful any-call	harmful unsafe	unsafe given call	benign any-call	benign over-refusal (judge)
Mistral-7B	0 → 11	0.39 → **0.02**	0.27 → 0.01	0.70 → 0.25 (n=4)	0.51 → **0.05**	0.03 → 0.08
Llama-3.1-70B	0 → 20	0.77 → **0.03**	0.52 → 0.02	0.67 → 0.80	0.93 → **0.12**	0.03 → 0.41
Gemma-3-27B	0 → 12162	0.15 → **0.04**	0.12 → 0.04	0.77 → 0.78	0.39 → **0.07**	0.22 → 0.14
Qwen3-14B	0 → 450	0.61 → 0.32	0.54 → 0.24	0.88 → 0.76	0.66 → 0.48	0.00 → 0.08
Command-R-7B	0 → 49	0.12 → 0.04	0.12 → 0.04	0.96 → 0.88	0.42 → **0.40**	0.03 → 0.18

Three things follow. Text-judged over-refusal understates the collateral damage badly — Gemma's *falls* while its benign task completion drops from 0.39 to 0.07, because a model that answers a tool task in prose is not refusing, but it is not doing the task either. Unsafe-given-call is flat or rising at every coefficient in every model: steering never makes the calls a model still makes safer. And Command-R at c=49 is the only point that cuts harmful unsafe calls (0.12→0.04) while keeping benign calls (0.42→0.40) — with unsafe-given-call still 0.88.

So the draft's "blunt lever" is, if anything, an understatement: the lever is engagement, and it moves benign and harmful engagement together. That is the same finding as ablation (§3.3) and patching (§3.1), from a third intervention. Report benign any-call next to over-refusal, and unsafe-given-call next to unsafe; `tab:steering-calls` in `paper_updates.tex` does both.

For Future Work: the target is a direction, or a different intervention, that changes unsafe-given-call rather than any-call. None of the three tried here does.

Separately, Gemma's published steering curve is flat for a fixable reason — it used Qwen's **absolute** grid (0/200/450/700) against a projection gap of 18,944, roughly 60× too small.

3.5 Prose that must change

Location	Problem
§5 ¶"A single direction causally governs text refusal"	Says ablation sharply decreases refusal "across every model". False for two of five.
§5 ¶"Tool context suppresses…"	"Text-derived and tool-derived directions remain aligned" — not true for Command-R or Llama.
§5 ¶"strong predictor but partial mediator"	AUC range 0.69–0.88 → 0.76–0.86. Patching flips are mostly "no call"; "partial mediator of engagement", not of tool-call safety. Patching numbers all change (§3.1).
§5 ¶"Steering is a blunt lever"	The 70 percent is degenerate output. The remaining evidence is call suppression on both sides (§3.4); report any-call and unsafe-given-call.
§5 Gemma, wherever ablation is described	No KL-admissible direction; raw-direction ablation is degenerate; under the mean-out recipe refusal *rises* 0.66→1.00. Say so with the KL stated (§1c).
§5 addition / "rescues Llama"	0→100% is degenerate output; at 1× the effect is $\leq$ 1% in Mistral and Gemma. Withdraw.
Methods, direction extraction	State that the KL guardrail was applied and which models fail it; state the render check and the degeneracy screen (§3.6).
§3.6 ¶"Robustness of the direction"	Promises a cosine analysis with no numbers; now has them.
§7 Discussion ¶"A shared refusal direction…"	"ablations collapse and decrease harmful prompt refusal rates" — needs softening.
Abstract + §1 Introduction	Still describe the mechanistic finding in terms of ablation collapsing refusal. Needs a pass once §5 settles.

`paper_updates.tex` has an outline for each of these — what the paragraph must cover and which numbers to cite — rather than drafted prose, so the writing stays in your voice. The three tables in that file are ready to paste as-is.

3.6 Methods additions (reproducibility)

Two things a reviewer could reasonably ask about, both outlined in `paper_updates.tex`:

- **Gemma tool prompts.** Gemma-3 has no tool-handling block in its chat template, in any published checkpoint. Its tool schemas have to be written into the prompt text. Validated: 12.5% unsafe against 14.6% from the behavioural run, and it reproduces the published AUC to within 0.001.
- **Llama weights.** Loaded from `unsloth/Meta-Llama-3.1-70B-Instruct` (the `meta-llama` repos are gated). Same bf16 weights; unlike another common mirror, its chat template carries tools.

---

4. Two problems found in the current draft

(a) Table 1 does not match the committed labels. Four of five text-refusal values agree to three decimals. Mistral reads **0.535** where summary_three_way.csv gives **0.435** — that looks like a transcription slip rather than a scoring difference.

(b) The unsafe/divergence columns run ~12% high. Every model's unsafe rate in Table 1 is above what the committed scorer produces (Qwen normal 0.544 vs 0.487; Gemma 0.172 vs 0.146), consistently. That suggests Table 1 was built with a stricter predicate that is not in the repo. This matters because the AUC column above uses the committed scorer — **if Table 1 keeps its current numbers, AUC should be regenerated against that same scorer so the two tables agree.** Whoever built Table 1 should confirm which scorer it used.

---

5. What is not settled

Patching is done (§3.1). What is left, in the order I would do it:

1. **Llama's numbers are from 4-bit NF4 weights.** The direction rebuilt from the 4-bit model has cosine 0.93 with the bf16 one, so it is a mild perturbation, but the patching and control rows for Llama should be confirmed in bf16 before they go in the paper. That needs a B200 or 2×H100 and ~30 minutes; the scripts take --load-4bit off and nothing else changes.
2. **Table 1 with your audited predicates.** The paper's unsafe columns are ~10 points above what our branch computes, which is almost certainly your add_call_columns + apply_fabricated_auth_overlay on main (now merged) rather than a bug. Recomputing is CPU only; it should land on the paper's numbers within a point or two. Mistral's refusal rate (0.435 here vs 0.535 in the draft) is still unexplained and may be the strip_tool_markup change; worth a look at the same time.
3. **The KL-admissible layer for Command-R (L27) and Llama (L60) differs from the operating layer.** Everything downstream — AUC, cosines, proj_gap, the patching layer — was computed at the operating layer. If the paper adopts the guardrail, it should adopt the selected layer consistently, which is one more GPU pass per model (~10 min each). The operating layers are within cosine 0.80–0.93 of the admissible ones, so I would expect small movements, not reversals.
4. **Suppression (Δ, t) is still carried over** from the pre-template-fix runs. Forward passes only; cheap to add to any session that has the models loaded.
5. **Gemma's tool-mode ablation** (§3.3) used the raw direction. If the mean-out direction is adopted for Gemma's no-tool ablation, the action-level experiment should use it too for consistency (~10 min).

None of these can revive the draft's Table 2. The narrower claim — the direction gates engagement, not action safety; Command-R is the one clean Arditi case; Qwen and Gemma refuse more when it is removed — is what the evidence supports, from three interventions and five models.

---

6. Bugs found (worth knowing, may affect other analyses)

- **Chat templates silently drop tools.** apply_chat_template(..., tools=...) ignores the tools when a template has no tool-handling block — no error. The tool-enabled prompt becomes a plain chat prompt and every response scores as safe. This hit Gemma-3 and one Llama mirror. Llama scored 0/100 unsafe at every steering coefficient before it was caught. _render_check.py asserts a tool name appears in the rendered prompt; run it before any tool-mode sweep.
- **Tool-call parsers must cover every family.** A parser handling only the Qwen/Mistral formats reads Llama's <|python_tag|> and Command-R's <|START_ACTION|> calls as "no tool calls", i.e. perfectly safe.
- **Over-refusal denominators.** Responses that are a bare tool call with no prose must count as *not refused* and stay in the denominator. Dropping them inflates every over-refusal rate.

---

7. Where things are

Branch	relabel-rerun (pushed; main merged in on 2026-09-02)
Tables to paste + writing outlines	paper_updates.tex (7 tables: mech, cosine, ablation-action, patching, steering-calls, controls, addition)
Predictions vs outcomes for the second session	PREDICTIONS.md
Full results write-up (first session)	relabel_analysis/RESULTS.md
Per-model numbers	relabel_analysis/gpu_*.json, steer_*.json, ablation_action_*.json, patching_*.json, controls_*.json (KL scan, diagnostics), controls_*_judged.json, steering_calls.json
Raw generations	relabel_analysis/steer_raw_*.json, patching_raw_*.json, controls_raw_*.json — every intervention output is saved; nothing needs regenerating to be rescored
Direction vectors	relabel_analysis/directions_*.pt; directions_controls_*.pt adds mean-out, random, and mean-direction vectors and mu_all
Regenerate the LaTeX	python3 make_paper_updates.py
Print tables	summarize_reruns.py, summarize_ablation_action.py, summarize_patching.py, summarize_steering.py
Rerun scripts	patching_rerun.py, controls_rerun.py (KL scan + controls + addition sweep), both with --load-4bit; judge_generations.py --raw-file judges any saved file
Degeneracy	degeneracy.py: is_degenerate (first audit) and is_degenerate_v2 (your rule; used for everything on 2026-09-02)
Pod logs	logs_pod/

First session: one B200 (183GB), all models in bf16. Second session (2026-09-02): one H100 80GB; Llama-70B in 4-bit NF4, marked wherever it appears.